

A Discussion on Acoustic Propagation in the Ice-covered Region of the Arctic Sea

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The results of a numerical investigation of the acoustic propagation in the ice-covered region of Arctic Sea where the environment can be considered range-independent are presented. Mathematical equations for Green's function and normal stress for a simple three layered model (vacuum-solid-fluid-fluid halfspace) are presented and investigated numerically. Environment parameters are motivated from measurements from Greenland Sea during Marginal Ice Zone Experiment, MIZEX [4]. This discussion is an effort to understand the acoustic plane propagation in the ice-covered region of the Arctic Sea.

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1. Introduction

In the recent decades, the enormous progress in computer science has stimulated acousticians to develop frequency dependent solutions for wave equation. In these numerically based techniques wave equation is solved by normal mode, fast field, and parabolic equation methods [3]. The choice of propagation model depends on the dominant characteristics of the environment and the frequency range that is under consideration. Hydrological measurements in the Marginal Ice Zone (MIZEX) of the Greenland sea reveal the need of a range-independent horizontal layered model in ice covered regions. SAFARI [2] provides accurate transmission loss values for propagation in range-independent multi-layered horizontally stratified environments and was therefore an effective tool for this investigation.

The work is structured as follows. First solution of wave equation for a range-independent environment is presented and basic equations for Green's function and normal stress are given for a three layered environment (vacuum-solid-fluid-fluid halfspace) in Sec. 2 and a numerical example (case 1) is presented to study a deep ice-covered sea, three other numerical examples study more complicated environments in Sec. 3. Even if the chosen simulated environments are very simple, remarkable results are observed. A summary ends the discussion in Sec. 4.

2. Governing Equations

By employing forward Fourier and Hankel transform respectively on the linear wave equation in a range independent environment, the depth-separated wave equation can be obtained [4]:

$$\left[\frac{\partial^2}{\partial z^2} - (k^2 - k_m^2(z))\right]G(k, z) = \frac{F(z)}{2\pi}. \quad (1)$$

Where k_m is medium wavenumber, k is the horizontal wavenumber and $F(z)$ is forcing term. The total solution of the depth-separated equation can be expressed as a superposition of a homogeneous solution G_H , satisfying boundary conditions, and a particular solution G_P , the free field produced by a source in an unbounded medium. $G_P(k, z)$ term is added to the homogeneous solution in layers containing one or more sources:

$$G(k, z) = G_H(k, z) + G_P(k, z). \quad (2)$$

$G_H(k, z)$ is found by employing the boundary conditions.

To find the total field at any range r and the time response, we apply the inverse Hankel transform.

$$g(r, z) = \int_0^\infty G(k, z) k J_0 dk. \quad (3)$$

where $J_0 = J_0(kr)$ is the Bessel function. The normal stress τ_{zz} is then given by:

$$\tau_{zz} = -\rho\omega^2 g(r, z). \quad (4)$$

where ρ the fluid density, ω the radial frequency and $g(r, z)$ is the potential field.

In this paper $G(k, z)$ and τ_{zz} are studied. The graphical results for the normal stress (NS) are given in [dB/1Pa] corresponding to the transmission loss definition, see [4], p. 67. To investigate the acoustic propagation in the ice-covered region a simple three layered model is considered (vacuum, solid, fluid, fluid half-space).

The integral representation of the homogeneous solutions in the layer 1 can be expressed as:

$$g_{11}(r, z) = \int_0^\infty [A_1^-(k)e^{-\alpha_1 z} + A_1^+(k)e^{\alpha_1 z}] k J_0 dk, \quad (5)$$

$$\text{where } \alpha_1 = \begin{cases} -\sqrt{k^2 - k_1^2}, & k > k_1, \\ i\sqrt{k^2 - k_1^2}, & k \leq k_1. \end{cases}$$

$$g_{s1}(r, z) = \int_0^\infty [B_1^-(k)e^{-\beta_1 z} + B_1^+(k)e^{\beta_1 z}] k J_0 dk, \quad (6)$$

$$\text{where } \beta_1 = \begin{cases} -\sqrt{k^2 - k_s^2}, & k > k_s, \\ i\sqrt{k^2 - k_s^2}, & k \leq k_s. \end{cases}$$

which can be explained as a plane wave decomposition of the field, where $A_1^-(k)$ and $A_1^+(k)$ are the amplitudes of the upgoing and downgoing compressional (longitudinal) field for the layer 1 and $B_1^-(k)$ and $B_1^+(k)$ are the amplitudes of the upgoing and downgoing shear field in the layer 1, $g_{11}(r, z)$ and $g_{s1}(r, z)$ are the potential fields in the solid corresponding to the compressional velocity and the shear velocity respectively. By using the Sommerfeld radiation condition, the total solution in the layer 2, where the source is located, can be expressed in the form of:

$$g_2(r, z) = \int_0^\infty \{[A_2^-(k)e^{-\alpha_2(z-D)} + A_2^+(k)e^{\alpha_2(z-D)}] + G_{P2}(k, z, z_s)\} k J_0 dk, \quad (7)$$

$$\text{where } \alpha_2 = \begin{cases} -\sqrt{k^2 - k_2^2}, & k > k_2, \\ i\sqrt{k^2 - k_2^2}, & k \leq k_2. \end{cases}$$

and $A_2^-(k)$ and $A_2^+(k)$ are the amplitudes of the upgoing and downgoing fields in the layer 2,

$G_{P2}(k, z, z_s) = \frac{S_\omega e^{\alpha_2 |z - z_s|}}{4\pi\alpha_2}$, S_ω is the source strength and D the thickness of the layer 1. Similarly the integral representation of the solution in layer 3 can be introduced as:

$$g_3(r, z) = \int_0^\infty A_3^+(k) e^{\alpha_3 [z - (D+L)]} k J_0 dk,$$

$$\text{where } \alpha_3 = \begin{cases} -\sqrt{k^2 - k_3^2}, & k > k_3, \\ i\sqrt{k^2 - k_3^2}, & k \leq k_3. \end{cases} \quad (8)$$

and A_3^+ is the amplitude of the downgoing field in layer 3 and L thickness of layer 2. The boundary conditions for such a model are presented in [4], p. 17. These lead to a linear equation system which can be written as matrix form with U as coefficient matrix and V as constant matrix:

$$U \begin{bmatrix} A_1^- \\ A_1^+ \\ B_1^- \\ B_1^+ \\ A_2^- \\ A_2^+ \\ A_3^+ \end{bmatrix} = V. \quad (9)$$

If the determinant of the coefficient matrix, $\text{Det}[U(k)]$, vanishes, then the solutions of the Eq. (9) have poles for the corresponding horizontal wavenumber values. The poles (pole) correspond to additional modes (mode) in the wavenumber spectrum. This phenomena can be observed in next part, Fig. 1 (a).

3. Numerical Examples

In this section we will study four cases to get an idea about sound propagation in the ice-covered region of the Arctic Sea. The measurements are motivated from MIZEX 87 [3] for which an ice-layer is included. In these calculations, the roughness and other inhomogeneous properties of ice are neglected.

3.1. Case 1: Three Layered Model, (vacuum-ice-fluid-fluid halfspace)

The first case considers an ice-covered sea environment for which a half-space fluid is included to neglect the effect of bottom properties. This model can be related to mathematical equations presented at previous section. The environmental parameters in the Table I were used to compute Green's function and normal stress (NS[dB/1Pa]) and results are shown in Fig. 1 and 2 for receiver depths 18 og 122m.

Table I. 3 layered waveguide (vacuum-solid-fluid-fluid)

	c_l	c_s	ρ	z
0 (vacuum)	0	0	0	∞
1 (ice)	$c_1 = 3000$	1600	0.92	$D = 4$
2 (water)	$c_2 = 1441$	0	1.0273	$L = 90$
3 (water)	$c_3 = 1457$	0	1.0279	∞

Where c_l [m/s] is the compressional sound velocity, c_s [m/s] is the shear sound velocity, ρ [gr/cm³] is density and z [m] is the layer depth and D denote ice thickness.

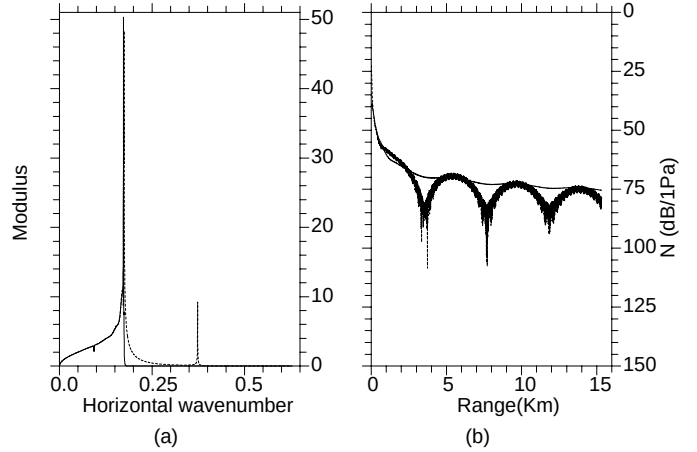


Figure 1: Results for Table I in layer 2 (RD=18m) and layer 3 (RD=122m) (a) modulus of depth-separated Green's function and (b) transmission loss for $f=40$ Hz, $N=4096$ (N: number of sampling points), $SD=7m$, $RD=18m$ (dashed lines) and $RD=122m$ (solid lines).

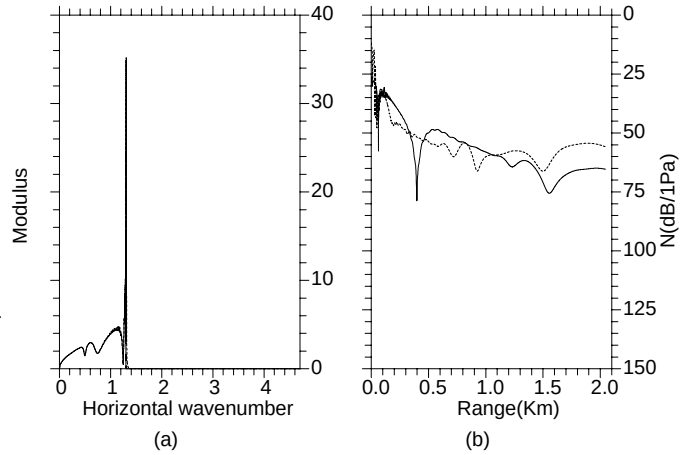


Figure 2: Results for Table I (a) modulus of depth-separated Green's function and (b) transmission loss for $f=300$ Hz, $N=16384$, $SD=7m$, $RD=18m$ (dashed lines) and $RD=122m$ (solid lines).

In Fig. 1 (a) ($f = 40$ Hz) the sharp peak is a discrete pole associated with the shear sound velocity in the ice.

In Fig. 1 (a) the last peak ($k \simeq 0.37 \text{ m}^{-1}$) is an additional pole which appears in the evanescent spectrum. It is a dramatic shear effect. The pole corresponds to vanishing of the $\text{Det}[U(k)]$, where U is given by Eq.(9). It belongs to the class of *leaky waves*. The peak disappears when the receiver is moved out deeper in the water ($RD=122m$), because in Eq.(7) the first exponential term becomes very small as the receiver depth becomes large.

In Fig. 2 (a) ($f = 300$ Hz) the sharp peak associated with sound propagation in the ice (shear velocity) is a discrete pole. The peak at $k \simeq 0.6 \text{ m}^{-1}$ is a part of the continuous spectrum, indicating a leaky mode associated with the compressional sound velocity in the ice.

In Fig. 1 (b) for $z = 18m$ the transmitted field in the layer 1 from layer 2 can be both compressional and shear fields and these can be reflected from the ice-vacuum interface and can be transmitted again as compressional field in the layer 2. The interference pattern has been dominated by the evanescent part.

3.2. Case 2: Three Layered Model (Vacuum, Arctic sound velocity profile)

The second case concerns a simple Arctic waveguide without ice layer.

Table II. Arctic waveguide 1.

	c_{l1}	c_{l2}	ρ	z
0 (vacuum)	0	0	0	∞
1 (water)	1439	1441	1.0273	80
2 (water)	1441	1456	1.0278	60
3 (water)	1456	1461	1.0279	360

Where c_{l1} is the compressional sound velocity at the top of the layer and c_{l2} compressional sound velocity at the bottom of the layer which is treated by SAFARI as a fluid with $\frac{1}{c_l^2(z)}$ varying linearly with depth. ρ is density.

The modulus of the Green's function and normal stress (NS [dB/1Pa]) for the environmental data given in Table II have been calculated in the layer 1 and 2 by SAFARI and shown in figures 3 and 4 for the frequencies 40 and 300 Hz respectively.

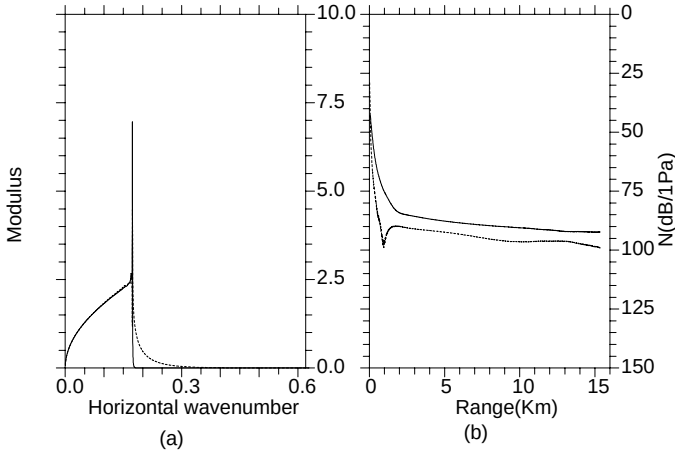


Figure 3: Arctic waveguide 1 (Table II), (a) modulus of depth-separated Green's function. (b) transmission loss for $f=40$ Hz, $N=4096$, $SD=3m$, $RD=18m$ (dashed lines) in layer 1 and $RD=122m$ (solid lines) in layer 2.

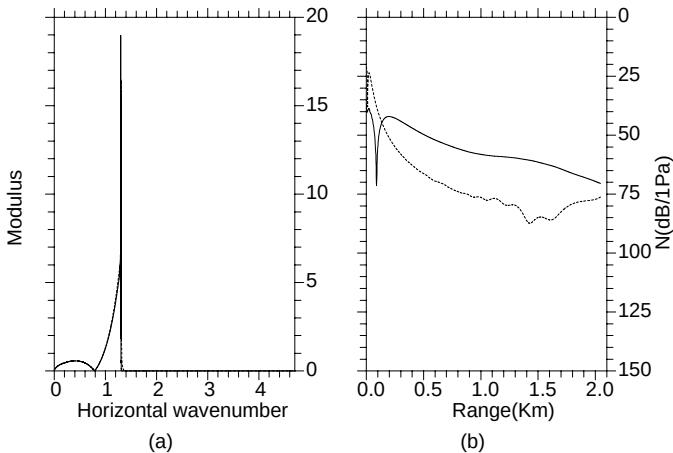


Figure 4: Arctic waveguide 1 (Table II), (a) modulus of depth-separated Green's function. (b) transmission loss for $f=300$ Hz, $N=16384$, $SD=3m$, $RD=18m$ (dashed lines) in layer 1, and $RD=122m$ (solid lines) in layer 2.

As one can observe evanescent field is important only for low frequencies at near field (in the number of wavelength), Fig. 3 (a).

Waves with frequencies lower than cutoff frequency $f_{cutoff} = \frac{1}{4L\sqrt{\frac{1}{c_1^2} - \frac{1}{c_2^2}}}$ can not propagate along the layer without damp-

ing. The cutoff frequency in layer 1 is about $f_{cutoff} \simeq 85$ Hz and in layer 2 about $f_{cutoff} \simeq 16$ Hz. Sound propagating with frequencies smaller than the cutoff frequency can transmit to the next layer and if the source frequency is larger than the cutoff frequency, then the sound energy can be channeled in that layer. In Fig. 3 (b) the sound energy leaks from layer 1 into layer 2 and is channeled and concentrated in the layer 2. Therefore the transmission loss at the receiver depth $z=122m$ is less than that of in the layer 1 ($z=18m$). This observation is in agreement with the ambient noise recording measurements made in Barents Sea during SIZEX 92 [1].

3.3. Case 3: Four Layered Model: Vacuum, Arctic sound velocity profile and bottom

The results of a study for the environmental data given in Table III, presented in case 4 (in which D , ice thickness, is zero) are shown in Fig. 5 and are compared with the results presented in Fig. 9 in Sec. 3.4.

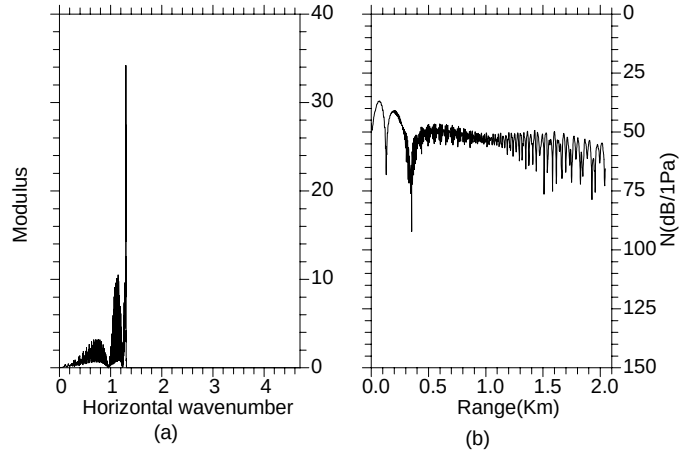


Figure 5: Results for Arctic waveguide 2 (Table II), for $D=0m$, (a) modulus of depth-separated Green's function in layer 3, $|G_3(k, z)|$, (b) transmission loss in layer 3 for $f=300$ Hz, $N=16384$, $SD=3m$ and $RD=122m$.

3.4. Case 4: Five Layered Model (Vacuum, ice, Arctic sound velocity profile and bottom)

Table III. Arctic waveguide 2.

	c_{l1}	c_{l2}	c_s	ρ	z (m)
0 (vacuum)	0	0	0	0	∞
1 (ice)	3000	3000	1600	0.92	D
2 (water)	1439	1441	0	1.0273	80
3 (water)	1441	1456	0	1.0278	60
4 (water)	1456	1461	0	1.0279	$360 - D$
5 (gravel)	1800	1800	0	2.0	∞

♣ A compressional attenuation ($\gamma_l = 0.6$ dB/ Λ) is applied for the absorbing bottom in the simulations.

The modulus of Green's function and normal stress for the environmental data given in Table III, with an ice thickness

$D = 4\text{ m}$, have been calculated and shown in figures and 6 for frequencies 300 Hz respectively.

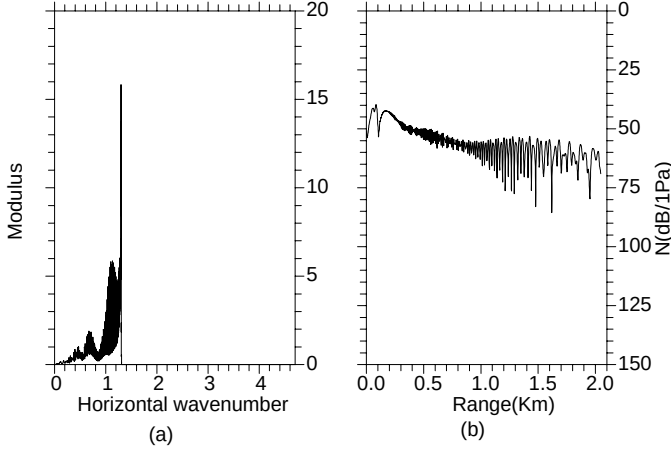


Figure 6: Results for Arctic waveguide 2 (Table III), for $D=4\text{m}$ (ice thickness), (a) modulus of depth-separated Green's function in layer 3, $|G_3(k, z)|$, (b) transmission loss in layer 3 for $f=300\text{ Hz}$, $N=16384$, $SD=7\text{m}$ and $RD=122\text{m}$.

Introduction of the ice has reduced the amplitude of the sharp peak. There are no significant changes in the corresponding normal stresses at the low frequencies comparing with the last model in which there was no ice at the top.

As frequency increases to $f = 300\text{ Hz}$, normal stresses shown in Fig. 6 (b), have been damped at short ranges, comparing with the case without ice layer Fig. 5 b). The reason is that the field can be transmitted in the ice for the short ranges. The transmitted field is mostly absorbed or trapped in the ice layer.

Comparing Fig. 1 with figures 5, 6 and 7 and Fig. 2 with figures 8 and 9, indicate that reflections from the sea bottom have significant effects on Green's function and normal stress.

4. Summary and Conclusions

The study is performed in a progression from simple environments to more complicated models that can simulate an ice-covered sea environment. Each model introduces greater complexity so that the importance of the various parameters can be isolated. The analytic integral expressions for the field for a simple 3 layered model are obtained by using boundary conditions. In these calculations only compressional monochromatic point sources have been used. In each experiment both the source and receivers are localized in the water below the ice interface.

The computations by SAFARI are performed for source frequencies at 40 and 300 Hz. The main observations and results from the numerical study are:

Sound will be transmitted into an ice layer as compressional and shear waves. These shear waves can be reflected from the ice-vacuum boundary and converted back to the compressional waves into water through the ice-water boundary. There is a competition between compressional and shear

waves at the ice-water boundary.

The subsequent shear waves are trapped in the ice and even a small shear attenuation eventually removes all the energy in the shear field. The trapped compressional waves in the ice layer are only important at short ranges.

A sensitivity study for a set of environmental models simulating the ice-covered region of the Arctic Sea is performed (not shown here) and the influence of bottom properties on transmission loss is studied and the results describe that the acoustic field propagating towards the bottom, will be reflected gradually and the propagation pattern is therefore less sensitive to bottom parameters specially at higher frequencies.

Acknowledgement

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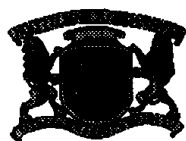
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